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Deep Power: Deep Learning Architectures for Power Quality Disturbances Classification

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***Abstract*—The transformation of the conventional electric power grid to modern smart grid are subjected to power system quality and reliability problems. In order to ensure reliable, secure and quality supply of power, it is important to characterize and classify the power quality disturbances. Power quality (PQ) disturbance classification schemes implicitly relies on feature engineering to extract unique and accurate features such as statistical information, spatio-temporal characteristics, stationary and non-stationary behavior of PQ signals. This paper explores the potentiality of deep learning algorithms to characterize and classify various PQ disturbances in smart grid. Deep learning algorithms have the inherent capability to automatically learn optimal features from raw input data and thus to avoid time-consuming feature engineering. To understand the effectiveness of various deep learning mechanisms, different architectures namely convolution neural network (CNN), recurrent neural network (RNN), identity-recurrent neural network (I-RNN), long short-term memory (LSTM), gated recurrent units (GRU) and convolutional neural network-long short-term memory (CNN-LSTM) are studied in this paper. Several experiments are conducted to propose an optimal deep learning architecture with specific network parameters and topologies. The performance of the proposed deep learning architecture is evaluated on a set of synthetic single and combined PQ disturbances and real-time PQ events. The proposed architecture is found to be accurate for real-time characterization and classification of power quality disturbances in smart grid.**

***Index Terms*—Smart grid, distributed generation system, deep power, power quality disturbances, deep learning, classification.**

I. Introduction

Micro grid or distributed generation (DG) system with variable renewable energy sources and energy conservation technologies are integrated in modern electric grid to satisfy growing power demand. Introduction of DG systems with multiple and non-linear load characteristics have brought great challenges to governing power quality [1], [2]. Increasing use of solid switching state devices, power electronic devices, use of non-linear loads with several environmental factors, and unbalanced power systems cause power quality (PQ) disturbances. Presence of power quality disturbances such as voltage sag, voltage swell, harmonics, inter-harmonics, transients, interruption, flicker, notch and their combinations result in deterioration of electric power quality. These disturbances cause malfunctioning of end-user equipments and enormous economic loss. Identification of PQ disturbances is fundamental to develop a compensation device for improving power quality. Conventional power analyzers fails to give sufficient knowledge on the disturbances [3]. The monitoring system must be able to detect and characterize PQ disturbances for reliable, secure and quality supply of power.

In literature, a significant amount of effort has been devoted to PQ disturbance analysis and classification [3]–[14]. Time-domain approaches, Fourier based techniques, Prony analysis, optimization algorithms such as Newton's method, least-square approaches, discrete wavelet transform (DWT) [4], empirical mode decomposition (EMD) [5], empirical wavelet transform [3], and variational mode decomposition (VMD) [6], [7] are studied to characterize PQ disturbances. The performance and accuracy of these methods greatly rely on the extracted features. The extracted, hand crafted features is fed as input to a shallow structured, pattern recognition technique, including hidden Markov model (HMM) [8], multilayer perceptron (MLP) [9], artificial neural network (ANN) [10], decision tree (DT) [11], logistic regression (LR) [12], support vector machine (SVM) [13] and extreme learning machine (ELM) [14]. It is inferred from literature that, accurate detection of PQ disturbances is dependant on signal processing techniques with predefined filter specifications, selection of unique and novel features, and an accurate pattern recognition method or classifier. Identification and extraction of hand crafted features require substantial effort. Even though aforementioned methods are successful, feature learning phase makes them time-consuming.

Recently, deep learning has become an emerging field, which have been impacted a wide range of tasks in artificial intelligence (AI) related to the field of signal, image and information processing [15]. Deep learning algorithms have multiple levels of representation and abstraction that help to make sense of data such as signals, images, and text. They have

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the immense capability to automatically learn optimal features from raw input data and thus to avoid feature engineering. Availability of large scale computing power such as general-purpose graphical processing units (GP-GPUs), significantly reduced hardware cost and the modern research advances in machine learning and related field have enabled the paradigm shift to deep learning techniques from conventional AI techniques [15], [16]. Through deep learning algorithms, machine can now surpass human interference in research areas of signal/information processing, pattern recognition, natural language processing, audio and video perception etc.

This paper explores the potentiality of deep learning architectures to characterize and classify various single and mixed PQ disturbances in modern smart grid. The proposed deep learning architecture automatically learns the best features without additional feature extraction stage and thus reduces the computation effort. Experiments are done with various network parameters and configurations to select an optimal architecture for PQ disturbance classification.

II. Materials and Methods

A. Power Quality (PQ) Disturbances

Power quality disturbances comprise of short and long duration voltage variations and waveform distortions. Voltage sag, swell and interruption belong to short duration PQ disturbances. Over voltage and under voltage conditions are termed as long duration voltage variations. Waveform distortions include steady state variations such as harmonics, inter harmonics, spikes, notches and noises. PQ disturbance signals have time-varying amplitudes. The performance of proposed deep learning architecture is validated on synthetic and real-time power quality disturbance signals.

B. Deep Learning Algorithms

Deep learning has become an important approach in artificial intelligence, signal/information processing and machine learning. Conventional machine learning tasks greatly dependent on extracted features, which need significant effort to find the most suitable features. Feature learning phase is task-specific or task-dependent while deep learning methods are data dependent. As the amount of data and range of application in AI tasks continues to grow, it becomes unavoidable to automatically learn features [17]. In deep learning techniques, the initial data fed into the deep network are sufficient to learn features and to classify the instances. A general deep learning architecture has several deep layers to obtain rich features of input data. The output from each previous layer is passed as input to successor layer, and learn multiple levels of data abstraction. A simple deep learning network has an input layer, multiple hidden layers followed by an output layer. Recently, a PQ signal classification scheme using CNN is proposed in [18]. This paper has reported an accuracy of 97% for classification of six different PQ classes. In [19], a stacked autoencoder framework for PQ disturbance classification is proposed. In this work, particle swarm optimization is employed to assist the classification. Indeed, the use of optimization approaches results additional computational complexity. Classification by using the image files of PQ events are proposed in [20].

To understand and evaluate various deep learning algorithms for PQ disturbance classification, different architectures such as convolution neural network (CNN) [21], sequential data modeling algorithms namely recurrent neural network (RNN) [22], identity-recurrent neural network (I-RNN) [23], long short-term memory (LSTM) [24], gated recurrent units (GRU) [25] and a hybrid architecture of CNN-LSTM are used in this paper.

1) Convolutional Neural Network (CNN): CNN architecture consists of a number of convolutional layers, subsampling layers and fully connected layers. The filters in convolutional layers produce an initial feature map of the input data. The convolution is performed by sliding the filter over the input data. Pooling layer or subsampling layer follows immediately after convolutional layer helps to simplify the information from convolutional layer. The feature map generated after pooling operation is then given to a fully connected layer or dense layer for classification. In the dense layer, a sigmoid non-linearity activation function is applied over the features for predicting the output class labels [26].

2) Recurrent Neural Network (RNN): RNN are extensions of feed-forward networks. In RNN, the output from one state is taken back as an input/feedback to it through a loop structure. RNN handles data in a sequential form and holds information in the network as a short memory through feedback loop. Thus in a given time instant, the hidden state holds information from all its previous states. The feedback loop makes RNN as extensions of feed-forward network which passes the information only in one direction. The hidden states act as memory units in the network, receives input from previous states and transfers output to next state. The hidden state vector h_t at a given time t is dependent on the current input x_t and previous state output h_{t-1}. That is,

$$h_t = f(Px_t + Wh_{t-1}) \tag{1}$$

where f represents the non-linear activation function like sigmoid, $tanh$, ReLU (rectified linear units). The output state vector y_t at time t is calculated from the hidden states at time t as,

$$y_t = soft\max(Qh_t) \tag{2}$$

In RNN, the hidden states are initialized randomly or as zeros. The main obstacle for RNN is the vanishing or exploding gradient problem, where the gradient vector explodes or decay over time steps. Hence, RNN fails to capture long-term dependencies.

3) Identity-Recurrent Neural Network (I-RNN): I-RNN is an extension to RNN architecture, where the initialization of weight matrix differs from traditional RNN. The weight matrix is initialized as identity matrix or its scaled versions. This initialization trick helps to avoid addition of error-derivatives during back propagation and thus enable long-term dependencies. The non-linear activation function used is ReLU.

4) Long Short-Term Memory (LSTM): LSTM a variant of RNN with a gating architecture, captures long-term temporal dependencies and avoid vanishing/exploding gradient problem. The LSTM blocks are known as memory cells with adaptive multiplicative gates namely input, output and forget gates. The input gate decides how much current information need to be passed.

$$i_t = \sigma(P^i x_t + W^i h_{t-1}) \tag{3}$$

The forget gate decides about the information need to be passed from previous state and is defined as,

$$f_t = \sigma(P^f x_t + W^f h_{t-1}) \tag{4}$$

The output gate defines the internal state information need to be passed.

$$o_t = \sigma(P^o x_t + W^o h_{t-1}) \tag{5}$$

The internal memory of the unit c_t is updated using previous memory c_{t-1} with forget gate and updated hidden state $\hat{h}$ with input gate.

$$c_t = c_{t-1} \circ f_t + \hat{h} \circ i_t \tag{6}$$

where $\circ$ denotes an element-wise multiplication operation, $\hat{h}$ is the candidate hidden state computed as,

$$\hat{h} = \tanh(P^h x_t + W^h h_{t-1}) \tag{7}$$

Now using the memory c_t, the output hidden state h_t is computed by multiplying with the output gate o

$$h_t = \tanh(c_t) \circ o_t \tag{8}$$

The long-term learning of temporal features is achieved through the gating mechanism in LSTM [27], [28].

5) Gated Recurrent Units (GRU): Similar to LSTM, GRU has gating units to pass information inside the unit. This architecture have two types of gates namely reset gate and update gate. The reset gate defines combination of the new input with memory from previous state and update gate defines how much information from previous state need to be preserved. At time instant t, these gate vectors are calculated as,

- Reset gate:

$$r_t = \sigma(P^r x_t + W^r h_{t-1}) \tag{9}$$

- Update gate:

$$u_t = \sigma(P^u x_t + W^u h_{t-1}) \tag{10}$$

The GRU architecture does not possess internal memory unit c_t and output gate o as in LSTM. In real-time learning, GRU architecture has fewer tuning parameters than LSTM.

C. Proposed Architecture - Convolution Neural Network-Long Short-Term Memory (CNN-LSTM)

In this paper, to improve the performance of PQ disturbance classification by learning additional features, a hybrid architecture is proposed. Fig. 1 represents various stages involved in the proposed hybrid architecture. The architecture contains an initial layer of CNN followed by max pooling to extract low level spatial dependencies among the data. To generate a more abstracted feature map, the first level extracted features are fed into the second layer of CNN and max pooling layers. The first CNN layer has 64 filters with a filter length of 3 for convolving with the input raw data. The second CNN layer contains 128 filters with the same filter length as in first CNN layer. The max pooling layer has used a standard size of 2, which combines the features from previous layer. The consolidated feature matrix generated from max pooling layer are then fed to an LSTM layer containing memory blocks. The LSTM layer extracts the long term temporal dependencies and generate a more abstracted feature map. The LSTM layer used in the proposed architecture has 50 memory blocks including several gates and activation function. Finally, the output of LSTM layer has passed to a fully connected layer or dense layer. Generally, a normal flat feed-forward neural network layer acts as fully connected layer, which contains units corresponds to the number of PQ classes. This layer has a non-linear activation function namely softmax function to calculate the probabilities of output class predictions. In this study, we have used a batch size of 32 and the model was trained for 1000 epochs.

D. Hyper Parameter Tuning

In general, deep learning architectures are parameterized models. Therefore the superior predictive performance of the

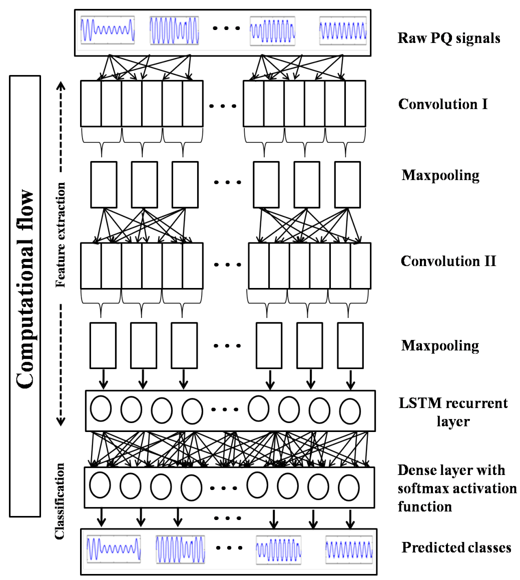

Fig. 1. Brief view of the stages involved in proposed hybrid (CNN-LSTM) architecture for PQ disturbance classification

models are dependant on the selection of optimal parameters and is achieved through hyper parameter tuning. Deep network parameters such as learning rate and number of memory blocks or number of filters have to be selected optimally to achieve the highest performance. In order to fix the hyper parameters a moderately sized RNN/LSTM/GRU/I-RNN/CNN networks which contains an input layer, hidden recurrent layer and an output layer is used. This initial network has an input layer with 1280 neurons, a single hidden recurrent layer/convolution layer followed by max-pooling and an output layer contains 11 neurons. The experiments are performed epoch-wise and we ran every experiment till 100 epochs. To find the suitable learning rate, we performed two trails of experiment for all deep learning architectures in the range of [0.01-0.5]. The highest performance is achieved at a learning rate of 0.01, after that the performance gradually decreased. Hence, in this study, the learning rate is fixed at 0.01. The performance of sequential data modeling algorithms such as RNN, I-RNN, LSTM and GRU depends on the number of memory blocks or hidden units. To fix the optimal number of memory blocks, two trails of experiments are done with hidden units/memory blocks as 32, 64, 128, 256, 512, 1024 and 2048. The highest performance is achieved with 1024 hidden units. Hence, the number of hidden units is taken as 1024 in this study. For CNN, the performance is relied on the number of filters and filter length. We performed two trails of experiment with CNN and max-pooling layer by varying the number of filters as 32, 64, 128 and filter length as 2, 3 and 6. The CNN model with number of filters as 64, each filter having a length of 3 has attained the highest accuracy and hence these values are fixed throughout in this study. After fixing the network parameters, experiments are conducted to select the most suitable network architecture for all deep learning models. Experiments are conducted with the following network topologies.

- RNN/LSTM/GRU/I-RNN - 1 layer with 1024 hidden units/memory blocks
- RNN/LSTM/GRU/I-RNN - 5 layer with 1024 hidden units/memory blocks
- RNN/LSTM/GRU/I-RNN - 10 layer with 1024 hidden units/memory blocks
- CNN - 1 layer
- CNN - 2 layer
- CNN - 1 layer followed by LSTM with 50 memory blocks
- CNN - 2 layer followed by LSTM with 50 memory blocks

From the experiments, the best performance is obtained for RNN/LSTM/GRU/I-RNN - 10 layer with 1024 units/memory blocks, CNN with 2 layers and CNN - 2 layer followed by LSTM with 50 memory blocks. For classifying the PQ signals, these best performed network topologies with optimal network parameters are used.

III. Results and Discussion

The platform chosen for performing the evaluation is Googles open source data flow engine TensorFlow [29] in conjunction with Keras [30]. To accelerate the performance and computations, the experiments are realized using back-propogation through time (BPTT) with adam optimizer [31] on GPU enabled TensorFlow in single NVidia GK110BGL Tesla k40. The evaluations are performed on real and synthetic PQ disturbances, for six deep learning architectures namely CNN, RNN, I-RNN, LSTM, GRU and hybrid model. In order to prevent overfitting and to improve the training a dropout parameter is used. The performance of the trained model of each deep learning architecture is evaluated using the test samples on epoch-wise. All the models have trained with a loss of categorical cross-entropy and with the hyper parameters explained in the previous section. The evaluations of the model are analyzed using metrics such as precision, recall, F1-measure and accuracy. The synthetic PQ disturbance signals

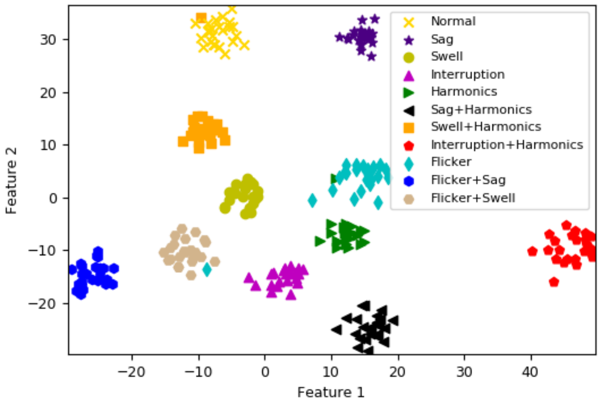

Fig. 2. Visualization of active values in final hidden layer using t-SNE

including waveform distortions and voltage fluctuations are generated in MATLAB using parametric equations [32]. These signals belong to 11 classes with single PQ disturbances such as voltage sag, voltage swell, harmonics, interruption, flicker, transients and combined PQ disturbances namely sag with harmonics, swell with harmonics, harmonics with interruptions, flicker with sag and flicker with swell. The signals are generated for 200 ms with a sampling rate of 6.4 kHz and each class have 250 signals. 60% of the data is randomly taken for training and the remaining 40% has given for testing. The raw PQ signals are passed through more than one hidden layers. The activation function in each hidden layer facilitates to distinguish the signals into different categories. Features learned in each layers are passed to next layer and finally the activation values in the last layer is able to separate the data into different classes. Each hidden layer includes high dimensional activation values. These high dimensional data are transformed to lower dimensional using t-SNE [33], as shown in Fig. 2. It is clear from Fig. 2, the deep learning model has learnt the features accurately and able to separate the classes efficiently. The results obtained for synthetic PQ signals are tabulated in Table I. The RNN model has attained an accuracy of 0.915 with a loss of 0.48. The I-RNN model accelerate the performance of conventional RNN and hence

TABLE I
THE PERFORMANCE OF DEEP LEARNING ALGORITHMS FOR PQ CLASSIFICATIONS

Classes	CNN		RNN		I-RNN		LSTM		GRU		CNN-LSTM	
	TPR	FPR	TPR	FPR	TPR	FPR	TPR	FPR	TPR	FPR	TPR	FPR
Normal	1.0	0.002	1	0	1	0	1	0.002	1	0	1	1
Sag	0.92	0.002	0.8	0.012	0.9	0.01	0.92	0.002	0.9	0.002	0.96	0.006
Swell	1	0	0.96	0.028	1	0.018	0.98	0.004	1	0.004	1	0
Interruption	0.96	0.008	1	0.016	1	0.012	1	0.01	1	0.012	0.98	0.004
Harmonics	1	0.002	1	0.006	1	0.004	1	0	1	0.002	1	0
Sag+ Harmonics	0.94	0.002	0.92	0	0.9	0.002	0.9	0	0.92	0.002	0.92	1
Swell+ Harmonics	1	0	0.96	0	0.96	0	1	0	0.98	0	1	0
Interruption+ Harmonics	0.98	0.004	0.98	0.006	0.98	0.01	0.98	0.01	0.96	0.008	1	0.008
Flicker	1	0	1	0.008	0.98	0.008	0.98	0.004	0.98	0.006	1	0
Flicker+ Sag	0.98	0.002	0.78	0.014	0.8	0.006	0.96	0.004	0.96	0.004	0.96	0
Flicker+ Swell	1	0	0.66	0.004	0.78	0	0.92	0	0.9	0	1	0
Accuracy	0.98		0.915		0.936		0.967		0.964		**0.984**	
Loss	0.18		0.48		0.42		0.30		0.34		**0.15**	
Precision	0.98		0.919		0.94		0.969		0.965		**0.984**	
Recall	0.98		0.915		0.936		0.967		0.964		**0.984**	
F1-score	0.98		0.912		0.935		0.967		0.964		**0.984**	

achieved an accuracy of 0.936. The recurrent LSTM and GRU has a higher performance with an accuracy of 0.967 and 0.964 respectively. The CNN architecture has got an accuracy 0.98 with a loss of 0.18. Hence, to improve the accuracy, in the proposed architecture, the features from CNN are given to a recurrent LSTM layer and achieved the highest accuracy of 0.984. The loss in classification is only 0.15.

The conventional PQ disturbance classification methods are time consuming mainly because of extraction of features. While the deep learning models has avoided the feature extraction stage and hence the processing time is comparatively smaller than conventional machine learning techniques. Additionally, the accuracy obtained for deep learning models are higher than conventional techniques. In [34], statistical feature based PQ classification scheme using support vector machine (SVM) and random kitchen sink (RKS) is discussed. The highest accuracy achieved for six PQ classes is only 94.44% with RKS classifier. On comparing with the results reported in [34], proposed deep learning architecture has obtained an accuracy of 98.4% with more PQ disturbances classes. This, in turn, highlight the scalability of deep learning models. That is, the deep learning models are efficient to learn distinguishable features across classes for accurate separation. In conventional approaches, as the number of classes increases, more sophisticated features are needed for efficient classification.

The real-time data samples taken from [35] containing three PQ disturbances namely voltage sag, transients and non-harmonic waveform distortions. A total of 1031 signals are taken for evaluation, from which 30% data are randomly taken for testing the model. The classification of real-time power quality events is performed using the hybrid architecture. The results of evaluation are tabulated in Table II. The CNN-LSTM architecture has achieved an accuracy of 0.919 with 0.47 loss.

TABLE II
THE PERFORMANCE OF CNN-LSTM HYBRID MODEL ON PQ SIGNALS FROM [35]

Classes	TPR	FPR
Class 1	0.686	0.007
Class 2	0.983	0.047
Class 3	0.950	0.931
Accuracy	**0.919**	
Loss	**0.47**	
Precision	**0.920**	
Recall	**0.919**	
F1-score	**0.916**	

The decreased performance of the algorithm is mainly due to the deficiency of data events for training the model. From the evaluation it is inferred that the performance of deep learning algorithms is proportional to the volume of data. To achieve a good separation between the PQ classes, a more abstracted, and higher level feature map is required.

A. Conclusion

This paper investigates the effectiveness of various deep learning architectures for PQ disturbances characterization and classification. The inherent mechanism of deep learning algorithms to automatically extract the best and unique features is utilized for the efficient and accurate classification. Deep learning models have scalability among data and are computationally efficient. To improve the performance of classification by learning additional features, this paper has proposed a hybrid architecture combining CNN with LSTM. In this hybrid architecture, initial two CNN layers are used for learning spatial information followed by a recurrent LSTM layer to learn the temporal characteristics. The features generated from maxpooling layer of CNN are fed to LSTM layer, followed by a dense layer with softmax activation function. The CNN-

LSTM hybrid architecture has obtained an accuracy of 0.984 on synthetic and 0.919 on real PQ signals. In addition, the performance of various deep learning architectures namely CNN, RNN, I-RNN, LSTM and GRU are studied. The efficiency of each algorithm is tested on synthetically generated and real-time PQ disturbance signals. This study serves as a bottom level system to understand the effectiveness of various deep learning approaches for PQ classification. Development of an intelligent deep learning PQ disturbance classification system by including more PQ disturbance classes is devised as a future scope of this work.

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